

Assignment 02

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Submitted To,

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Thermal, Microwave and hyperspectral remote sensing

Derivation of Thermal Image using LANDSAT 8/9 Data and estimation of Land Surface Temperature(LST) in QGIS

Gemini pro:

Prompt 1: What is a thermal image in remote sensing

In remote sensing, a **thermal image** (often called a **thermogram**) is a visual representation of the temperature of objects on the Earth's surface.¹

While a standard satellite image captures reflected sunlight (like a photo), a thermal image captures **emitted energy** in the form of heat.² This allows us to "see" things that are invisible to the naked eye, such as underground fires, heat escaping from buildings, or moisture in soil.³

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1. The Anatomy of a Thermal Image

A thermal image doesn't actually contain "color" in the traditional sense. It is a data-driven map where each pixel represents a specific temperature measurement.⁴

- **Greyscale Representation:** By default, raw thermal data is often displayed in greyscale.⁵
 - **Lighter/Whiter tones** indicate higher temperatures (more heat energy).⁶
 - **Darker/Black tones** indicate cooler temperatures.
- **Pseudo-Color (False Color):** To make images easier for humans to interpret, software applies a "color palette."⁷
 - **Example:** A "Ironbow" palette might map hot areas to yellow/white, medium heat to red/orange, and cold areas to dark blue or purple.

2. Key Characteristics

Thermal images have unique properties that distinguish them from the optical photos you see on Google Maps:

- **Day and Night Acquisition:** Since the sensor measures heat emitted by the Earth itself, it doesn't need the sun.⁸ A thermal image taken at midnight can look almost as clear as one taken at noon.
- **Coarser Resolution:** Because thermal radiation has longer wavelengths and lower energy than visible light, thermal sensors usually have a "lower resolution."⁹ A pixel in an optical image might represent 1 meter, while a thermal pixel might represent 30 to 100 meters.
- **Atmospheric Windows:** To get a clear image, satellites must look through "windows" in the atmosphere where water vapor doesn't block the heat.¹⁰ These are typically the 3–5 ¹¹\$\mu\$m and 8–14 ¹²\$\mu\$m ranges.¹³

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3. What Does the Image Actually Show?

It is important to understand that a thermal image shows **Radiant Temperature**, which can be slightly different from the actual physical (kinetic) temperature.

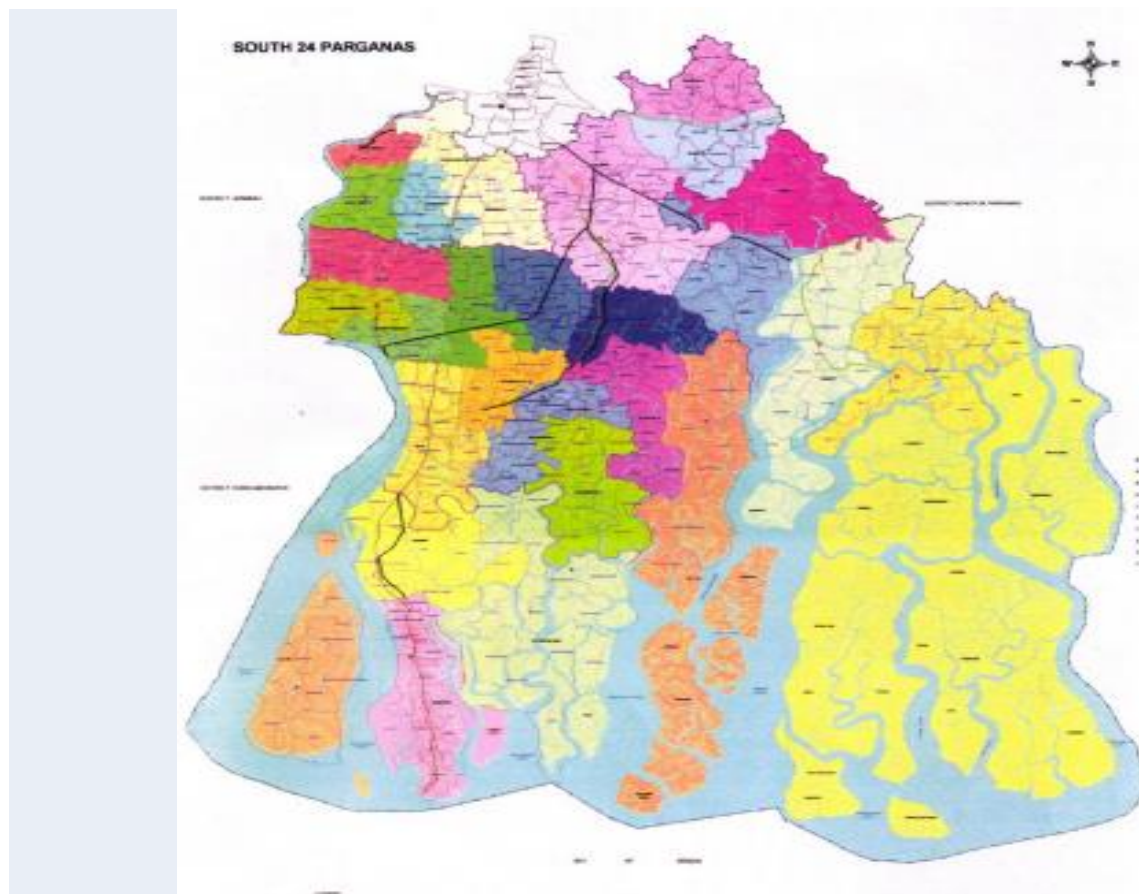
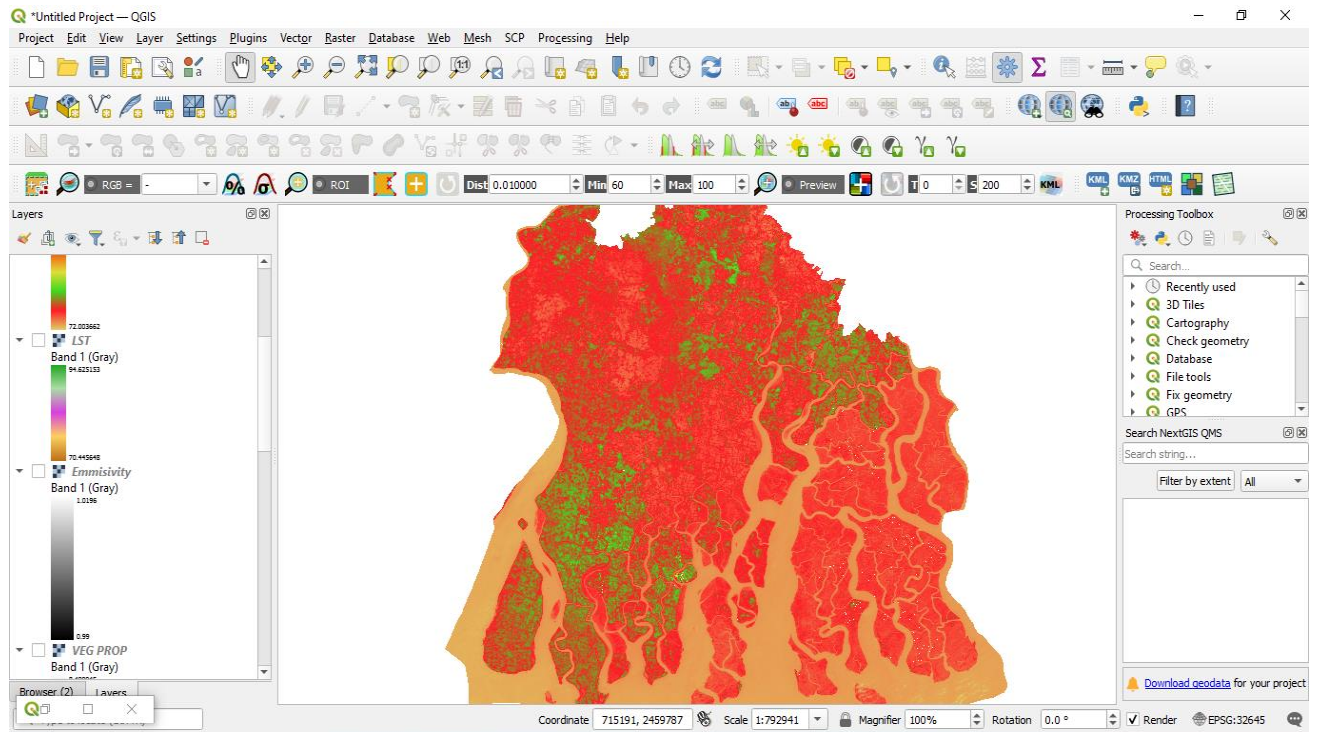
This is due to **Emissivity (ϵ)**. Different materials emit heat differently:

- **Water** is a "near-perfect" emitter ($\epsilon \approx 0.99$), so its thermal image is very accurate.
- **Polished Metals** have very low emissivity ($\epsilon \approx 0.10$). In a thermal image, a hot metal roof might look "cold" because it is reflecting the cold sky rather than emitting its own heat.

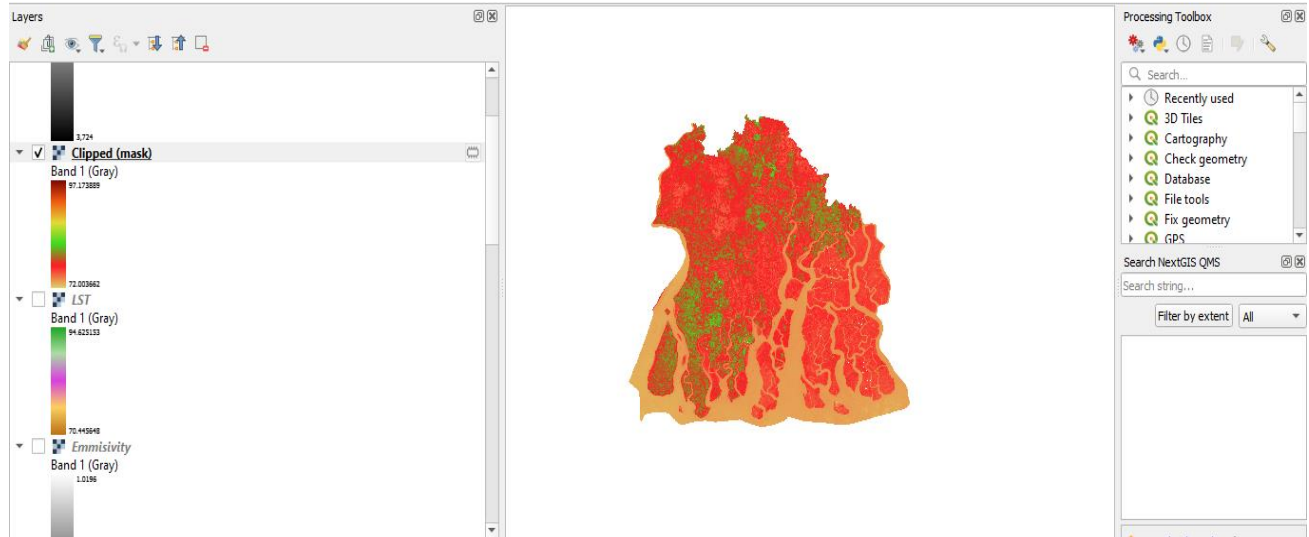
4. Comparison: Optical vs. Thermal

Feature	Optical Image	Thermal Image
Source	Reflected Sunlight	Emitted Heat
Primary Use	Mapping, Land Cover	Temperature, Energy, Moisture
Night Use	No (requires flash/lights)	Yes (excellent)
Best Feature	High Detail/Color	Detects invisible heat patterns

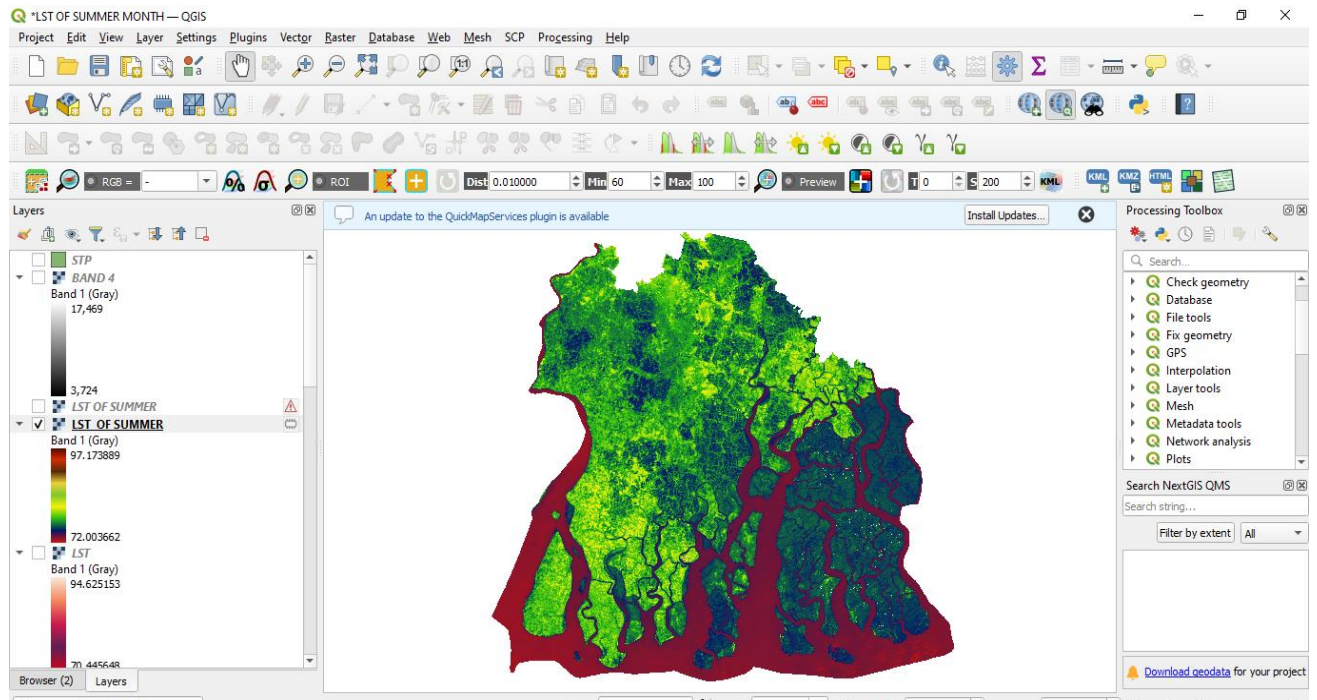
[Image comparing an optical satellite view of a forest and a thermal view showing a hidden fire front under the smoke]



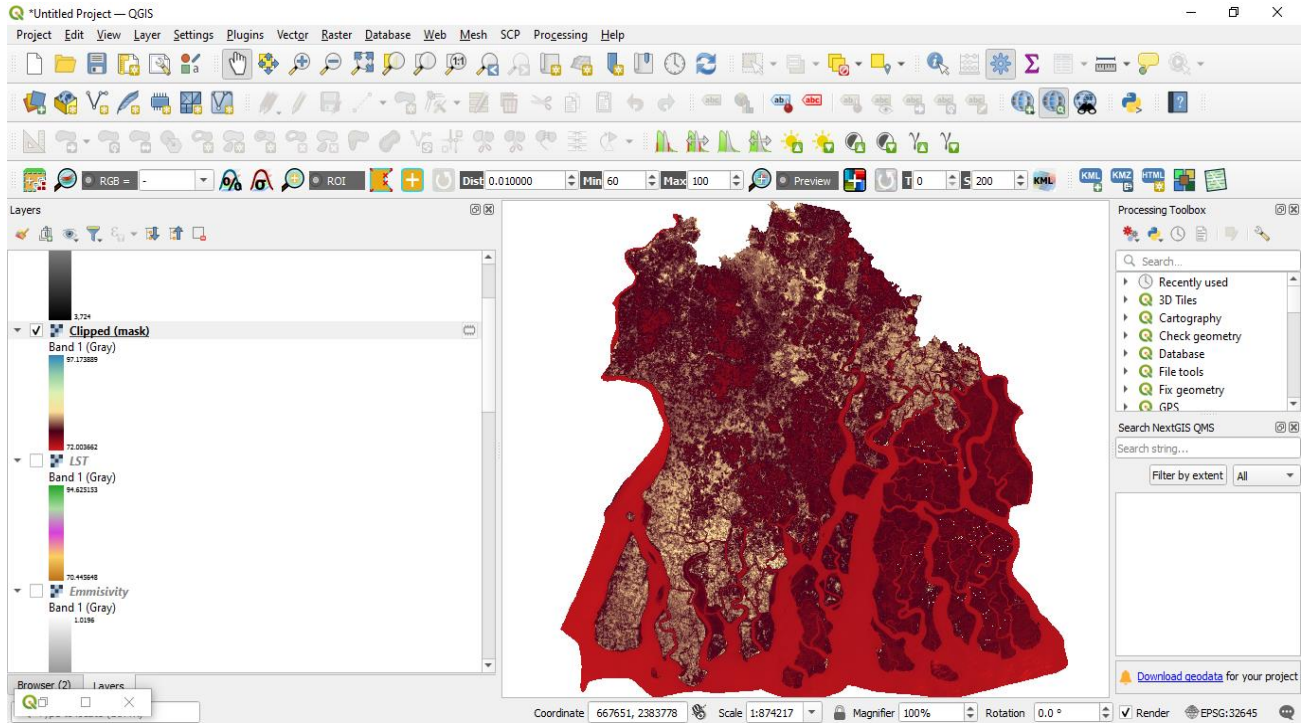
Map layout of SOUTH 24 PARGANAS



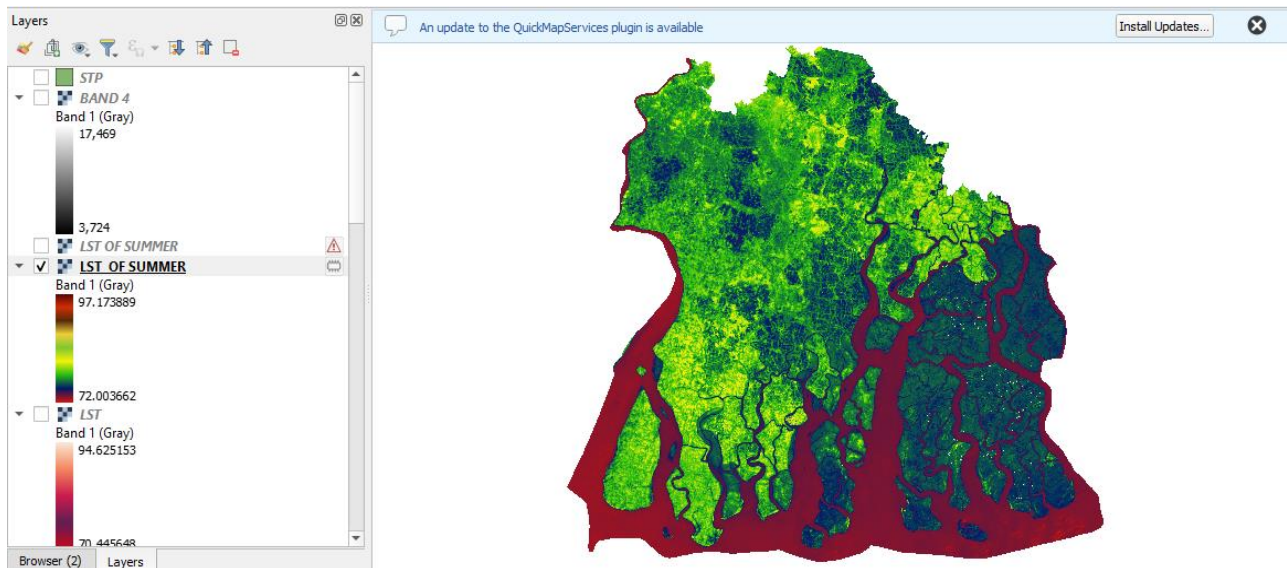
Import of district of SOUTH 24 PARGANAS Map to QGIS



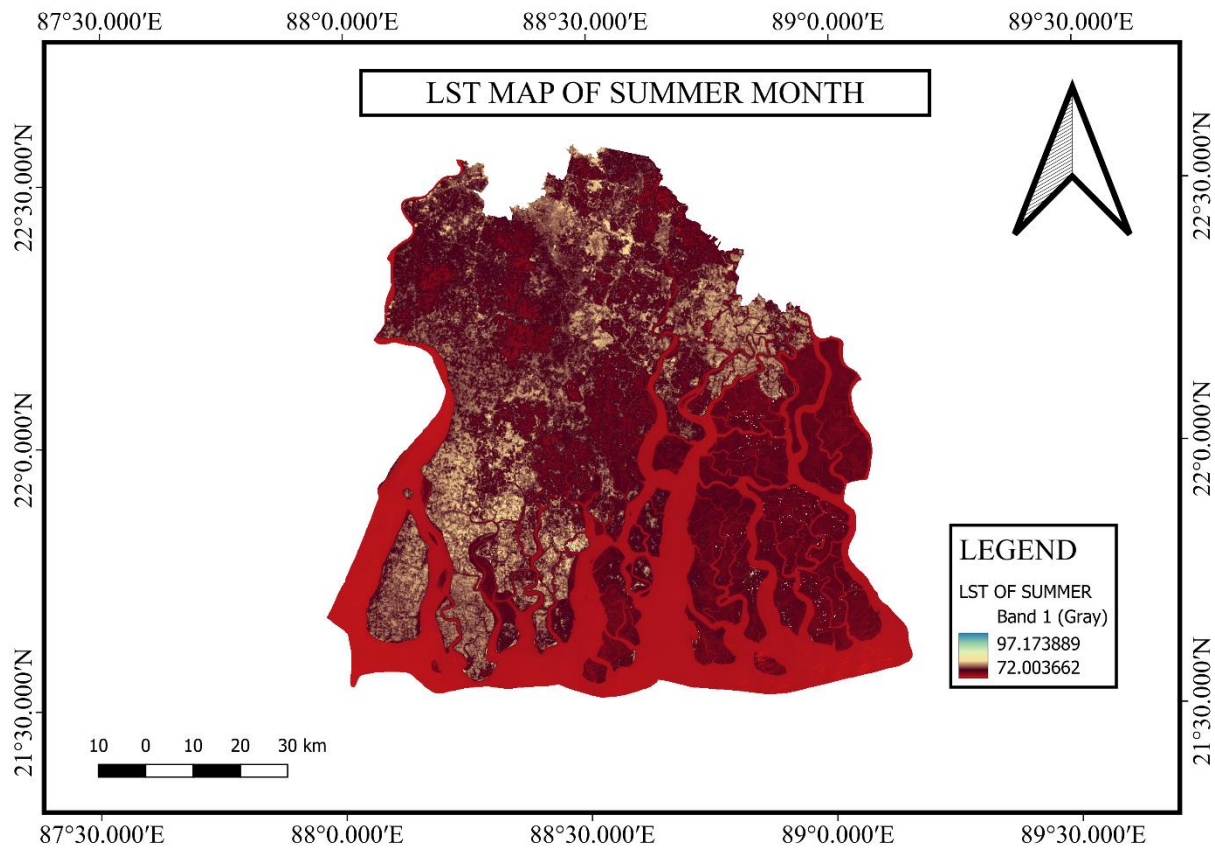
Changing the symbology of the LST Image



Import of district of SOUTH 24 PARGANAS Map to QGIS



Display of LST image and LST Value in summer



LST Map of Summer month

Prompt 2: How we can analyze LST for a given district according to seasonal changes (summer and winter)?

Analyzing the seasonal variation of Land Surface Temperature (LST) requires a systematic geospatial approach, typically using satellite imagery like **Landsat 8/9** or **MODIS**.¹ For a district like South 24 Parganas, the analysis involves comparing datasets from two distinct windows: **Summer (Pre-Monsoon: March–May)** and **Winter (January–February)**.

1. Data Selection & Tools

- **Satellite Data:** * **Landsat 8/9 (Level 2):** Best for detailed district-level study (30m resolution).
 - **MODIS (MOD11A2):** Better for observing daily/weekly trends over large areas, though at lower resolution (1km).²
- **Software:** You can use **Google Earth Engine (GEE)** for cloud-based processing or **ArcGIS/QGIS** for desktop-based manual analysis.³

2. Step-by-Step Analysis Workflow

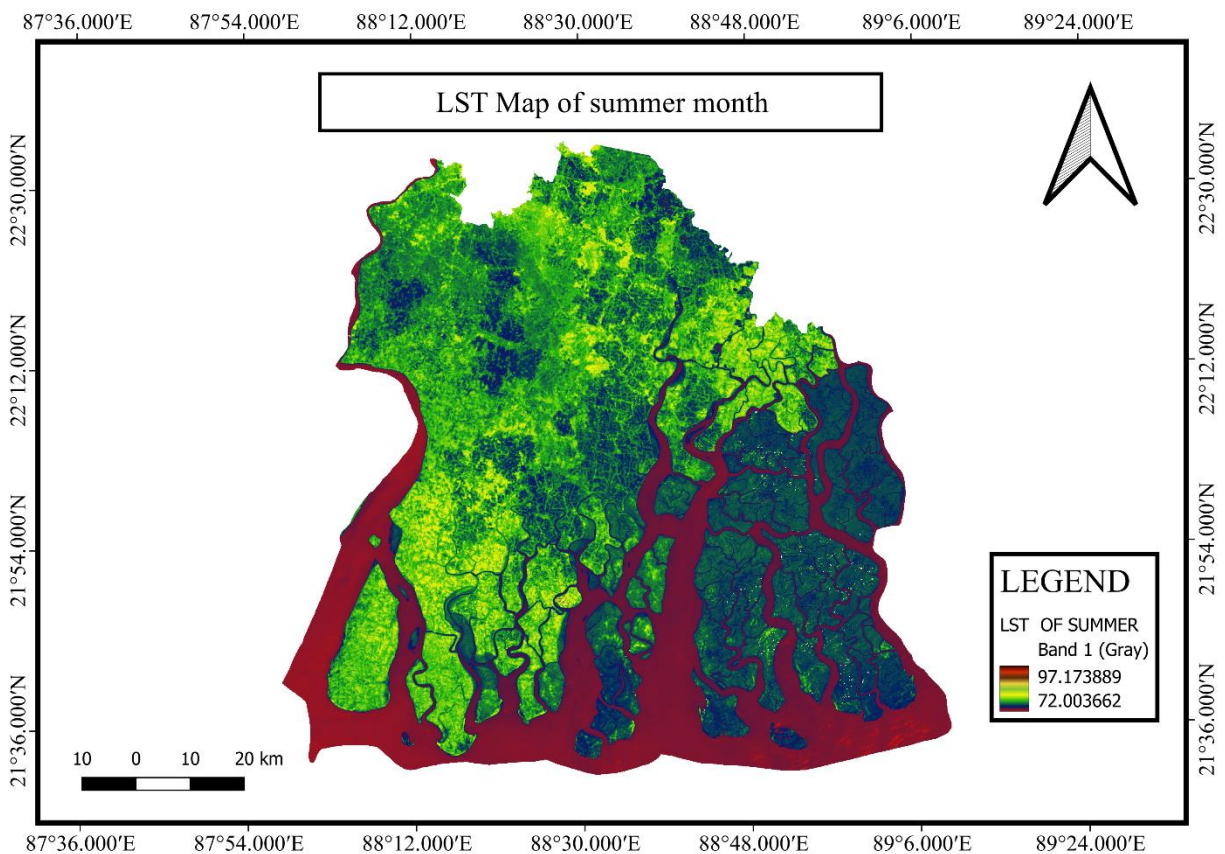
To get from a raw satellite image to a seasonal comparison map, follow these technical steps:

Phase A: Pre-Processing

1. **Image Selection:** Download or filter images with <10% cloud cover for the specific months of summer and winter.
2. **Radiometric Correction:** Convert Digital Numbers (DN) to **Top of Atmosphere (TOA) Radiance**.
 - o Note: If using Landsat Level 2 "Science Products," this is often pre-calculated.

Phase B: Calculating Indices

1. **Brightness Temperature (BT):** Convert the thermal band radiance into Kelvin using the satellite's thermal constants ($^4K_{\{1\}}$ and $^5K_{\{2\}}$).⁶
2. **NDVI (Vegetation Index):** Calculate $NDVI = \frac{(NIR - Red)}{(NIR + Red)}$. This is crucial because vegetation significantly cools the surface.
3. **Land Surface Emissivity (ϵ):** Use the NDVI to estimate how efficiently the surface emits heat.⁷ Water, forests, and concrete all have different "emissivity" values.



LST Map of South 24 Parganas in summer month

Phase C: The LST Formula

Apply the final LST equation to convert Brightness Temperature into actual Surface Temperature in Celsius:

$$LST = \frac{BT}{1 + \left(\lambda \cdot \frac{BT}{\rho} \right) \cdot \ln(\epsilon)}$$

Where λ is the wavelength of emitted radiance and $\rho = 1.438 \times 10^{-2} \text{ mK}$.

3. Interpreting the Comparison

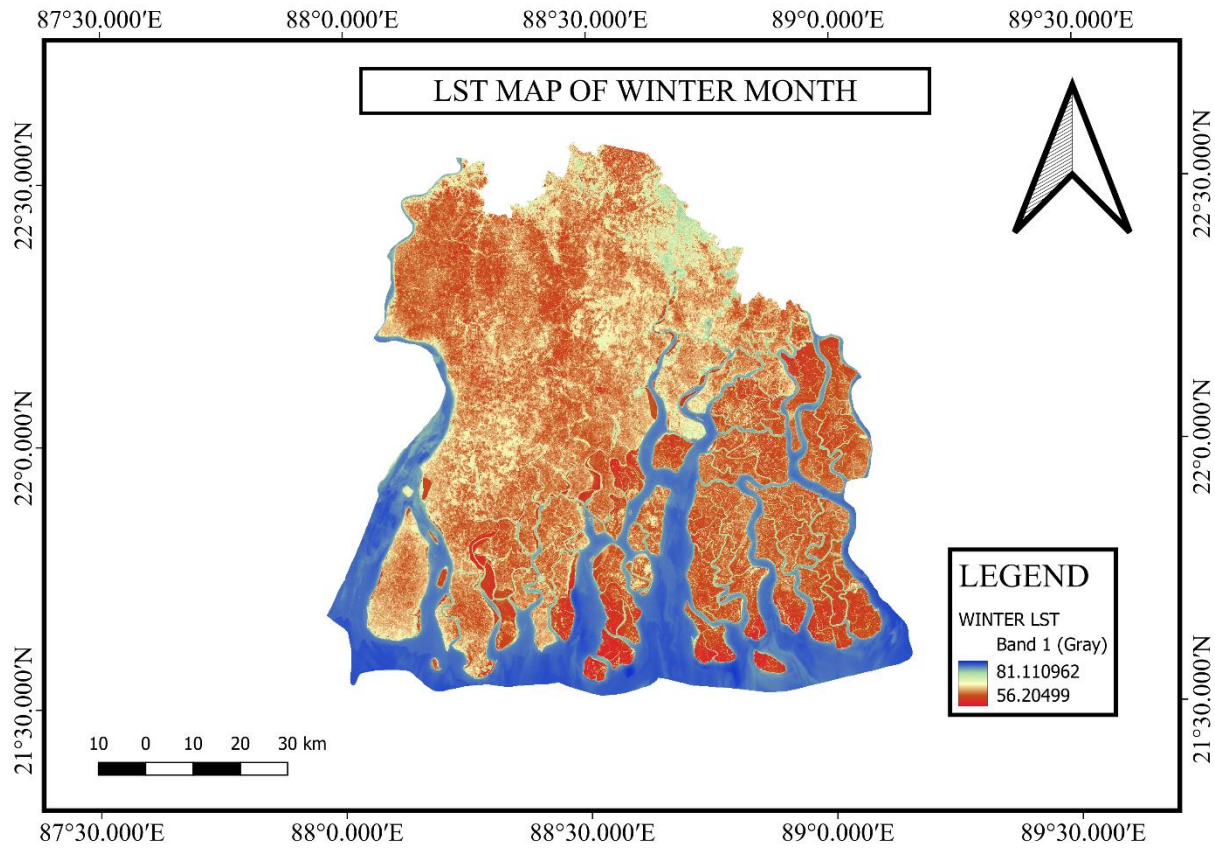
Once you have your two maps (Summer vs. Winter), look for these specific indicators:

- **ΔLST (Temperature Difference):** Create a "Difference Map" by subtracting the Winter LST from the Summer LST ($LST_{\text{sum}} - LST_{\text{win}}$). This highlights which parts of the district (like urban Baruipur) heat up the most.
- **LULC Correlation:** Overlay a **Land Use/Land Cover (LULC)** map. You will typically find:
 - **Urban/Built-up:** High LST in summer; retains heat longer in winter.¹¹
 - **Mangroves/Forests:** Stable, lower LST due to "Evapotranspirative cooling."
 - **Agricultural Land:** High LST in summer (dry fallow soil); low LST in winter (moist/cropped soil).

4. Why this matters for South 24 Parganas

In this specific district, the **Sundarbans** act as a natural "heat sink." Analyzing the seasonal LST helps in:

- **Disaster Management:** Identifying blocks prone to severe heatwaves.
- **Urban Planning:** Monitoring how the expansion of Kolkata's suburbs is creating new "Urban Heat Islands."
- **Ecology:** Tracking how rising temperatures might be stressing the mangrove ecosystem.



LST Map of South 24 Parganas of Winter Month

Prompt 3: Give the entire process and formulaes to calculate LST

Calculating Land Surface Temperature (LST) is a multi-step process that converts the raw Digital Numbers (DN) of satellite imagery into physical temperature units ($^{\circ}\text{C}$).

The following guide uses **Landsat 8/9** as the standard, as it is the most common data source for district-level analysis in South 24 Parganas.

Step 1: Top of Atmosphere (TOA) Spectral Radiance

The raw pixels (DN) must first be converted into spectral radiance (L_{λ}), which is the amount of radiation hitting the satellite sensor.

Formula:

$$L_{\lambda} = M_L \cdot Q_{cal} + A_L$$

- M_L : Band-specific multiplicative rescaling factor (from Metadata).

- A_L : Band-specific additive rescaling factor (from Metadata).
- Q_{cal} : The raw pixel value (Band 10).

Step 2: Top of Atmosphere Brightness Temperature (BT)

Brightness temperature is the temperature of a blackbody that would emit the same amount of radiation as observed by the satellite.

Formula:

$$BT = \frac{K_2}{\ln \left(\frac{K_1}{L_{\lambda}} + 1 \right)} - 273.15$$

- K_1 and K_2 : Thermal conversion constants (found in the MTL metadata file).
- **Result:** Temperature in Celsius ($^{\circ}\text{C}$).

Step 3: Normalized Difference Vegetation Index (NDVI)

LST is heavily influenced by vegetation. We need NDVI to calculate the "emissivity" (how well the surface releases heat) in the next step.

Formula:

$$NDVI = \frac{\text{Band 5 (NIR)} - \text{Band 4 (Red)}}{\text{Band 5 (NIR)} + \text{Band 4 (Red)}}$$

Step 4: Proportion of Vegetation (P_v)

This step estimates the density of vegetation in each pixel.

Formula:

$$P_v = \left(\frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}} \right)^2$$

- $NDVI_{max}$ and $NDVI_{min}$: The maximum and minimum values found in your NDVI map.

Step 5: Land Surface Emissivity (ϵ)

Emissivity is the ratio of energy radiated by a surface to energy radiated by a blackbody. Surfaces like concrete have different emissivity than a mangrove forest.

Formula:

$$\epsilon = 0.004 \cdot P_v + 0.986$$

- *Note: 0.986 is a standard correction value for land surfaces.*

Step 6: Final Land Surface Temperature (LST)

This final formula combines the Brightness Temperature and Emissivity to give the true "skin" temperature of the earth.

Formula:

$$LST = \frac{BT}{1 + \left(\lambda \cdot \frac{BT}{\rho} \right) \cdot \ln(\epsilon)}$$

- λ : Wavelength of emitted radiance (approx. **10.89** for Landsat 8 Band 10).
- ρ : A constant ($h \cdot c / \sigma$) equal to **14388** $\mu\text{m K}$.
- ϵ : Emissivity calculated in Step 5.

Summary Table for Landsat 8 (Band 10)

Parameter	Metadata Keyword / Value
Radiance Mult (M_L)	RADIANCE_MULT_BAND_10 (usually 0.0003342)
Radiance Add (A_L)	RADIANCE_ADD_BAND_10 (usually 0.1)
K_1 Constant	K1_CONSTANT_BAND_10 (approx. 774.88)
K_2 Constant	K2_CONSTANT_BAND_10 (approx. 1321.07)

Prompt 4: How is the variation of land surface temperature for south 24 parganas from summer to winter months

In South 24 Parganas, the Land Surface Temperature (LST) undergoes a significant seasonal shift, driven by the district's unique geography—ranging from the urbanized northern fringes near Kolkata to the vast mangrove forests and tidal networks of the Sundarbans in the south.

Unlike air temperature, **LST** (the "skin" temperature of the earth) is highly sensitive to land cover, such as concrete, water, or dense vegetation.¹

1. Summer (Pre-Monsoon): March to May

This is the period of peak thermal intensity. The LST reaches its maximum during these months due to high solar radiation and low soil moisture.

- **Temperature Range:** LST typically ranges from **32°C to 38°C**, though artificial surfaces (built-up areas) can soar above **40°C**.
- **Spatial Variation:**

- **Urban Hotspots:** Areas like Baruipur and the northern borders near Kolkata show the highest LST due to the **Urban Heat Island (UHI)** effect.
- **Agricultural Land:** Harvested or fallow fields act as "bare soil," which absorbs significant heat, often recording LST nearly as high as urban zones.
- **The Sundarbans:** The southern mangrove regions stay significantly cooler (often **4-6°C lower** than urban areas) due to dense canopy shading and high evapotranspiration.

2. Monsoon: June to September²

With the arrival of the rains, LST drops across the district and becomes more uniform.

- **Temperature Range:** Average LST settles between **28°C and 31°C**.
- **Impact of Moisture:** The high cloud cover reduces direct solar radiation, and increased soil moisture allows the land to dissipate heat more effectively through evaporation. Water bodies and wetlands (which are abundant in this district) maintain the lowest and most stable temperatures during this time.

3. Winter: November to February

Winter brings the most drastic cooling, with January being the coldest month.³

- **Temperature Range:** LST typically fluctuates between **18°C and 25°C** during the day, while nighttime "skin" temperatures can drop to **12°C–15°C**.
- **Seasonal Contrast:** * The difference between summer and winter LST in South 24 Parganas can be as much as **15°C to 18°C**.
 - **Inversion Effect:** During winter nights, concrete and paved surfaces in the urban north shed heat quickly, but "thermal inertia" can sometimes keep built-up areas slightly warmer than the surrounding rural paddy fields.

Comparison of Seasonal LST Characteristics:

Feature	Summer (May)	Winter (January)
Typical LST Range	34 [°] C - 39 [°] C	18 [°] C - 24 [°] C
Highest LST Zones	Built-up areas & Fallow land	Urban centers (Night heat retention)
Lowest LST Zones	Mangroves & Deep water	Rural wetlands & Shaded forests
Primary Driver	Solar Insolation	Atmospheric Cooling & Dry Air

Prompt 5: Google Earth engine code to calculate LST (for reference)

```
// 1. Define the Area of Interest (South 24 Parganas)
// Note: You can also upload a Shapefile to assets and call it here.
var roi = ee.FeatureCollection("FAO/GAUL/2015/level2")
    .filter(ee.Filter.eq('ADM2_NAME', 'South 24 Parganas'));

Map.centerObject(roi, 9);
Map.addLayer(roi, {color: 'grey'}, 'South 24 Parganas Boundary', false);

// 2. Load Landsat 8 Collection 2 Level 2 (Surface Reflectance & Temp)
// We filter for a specific seasonal window (e.g., Summer 2023)
var summer_collection = ee.ImageCollection("LANDSAT/LC08/C02/T1_L2")
    .filterBounds(roi)
    .filterDate('2023-03-01', '2023-05-31')
    .filter(ee.Filter.lt('CLOUD_COVER', 10))
    .median()
    .clip(roi);

// 3. Calculation of NDVI (Using Band 5 and Band 4)
var nir = summer_collection.select('SR_B5');
var red = summer_collection.select('SR_B4');
var ndvi = nir.subtract(red).divide(nir.add(red)).rename('NDVI');

// 4. Calculate Proportion of Vegetation (Pv)
var min = ee.Number(ndvi.reduceRegion({
    reducer: ee.Reducer.min(),
    geometry: roi,
    scale: 30,
```

```
maxPixels: 1e9
}).values().get(0));

var max = ee.Number(ndvi.reduceRegion({
  reducer: ee.Reducer.max(),
  geometry: roi,
  scale: 30,
  maxPixels: 1e9
}).values().get(0));

var pv = (ndvi.subtract(min).divide(max.subtract(min))).pow(ee.Image(2)).rename('Pv');

// 5. Calculate Land Surface Emissivity (E)
// Formula:  $0.004 * P_v + 0.986$ 
var emissivity = pv.multiply(0.004).add(0.986).rename('Emissivity');

// 6. Select Thermal Band and Convert to Brightness Temperature (BT)
// Landsat 8 Band 10 is the Thermal band.
// Collection 2 Level 2 thermal bands are already scaled.
// Scale factor is 0.00341802 and offset is 149.0 for Kelvin.
var thermal = summer_collection.select('ST_B10');
var bt = thermal.multiply(0.00341802).add(149.0); // Result in Kelvin
var btCelsius = bt.subtract(273.15).rename('BT_Celsius'); // Convert to Celsius

// 7. Calculate Final Land Surface Temperature (LST)
// Formula:  $LST = BT / (1 + (\lambda * BT / \rho) * \ln(\text{emissivity}))$ 
var lambda = 10.895; // Wavelength for Band 10
var rho = 14388; // constant  $h*c/\sigma$ 
```

```
var lst = btCelsius.divide(  
  btCelsius.multiply(lambda).divide(rho).multiply(emissivity.log()).add(1)  
).rename('LST_South24Parganas');
```

```
// 8. Visualization
```

```
var lstVis = {  
  min: 25,  
  max: 45,  
  palette: [  
    '0000ff', '00ffff', 'ffff00', 'ff0000', 'b22222'  
  ]  
};
```

```
Map.addLayer(lst, lstVis, 'Land Surface Temperature (Celsius)');
```

```
// 9. Export the result to Google Drive
```

```
Export.image.toDrive({  
  image: lst,  
  description: 'LST_South_24_Parganas_Summer',  
  scale: 30,  
  region: roi,  
  fileFormat: 'GeoTIFF',  
  maxPixels: 1e9  
});
```

How to use this script:

1. **Open GEE:** Go to code.earthengine.google.com.
2. **Paste & Run:** Paste the code into the editor and click **Run**.
3. **Winter Analysis:** To calculate the winter variation, change the date range in Step 2 to '2023-12-01' to '2024-01-31'.

4. **Legend:** The map will display a color gradient where **Blue** represents cooler surfaces (wetlands/mangroves) and **Dark Red** represents hot surfaces (urban/fallow land).

Key Variables Explained:

- **Scale Factors:** In Landsat Collection 2, the thermal data is pre-processed. We apply 0.00341802 to get Kelvin directly from the pixel value.
- **ROI:** The script uses the Global Administrative Unit Layer (GAUL) to automatically find the South 24 Parganas boundary.
- **Emissivity Correction:** This is the most critical part of LST. By calculating NDVI first, we tell the script exactly which pixels are dense forest (high emissivity) and which are bare soil (lower emissivity), making the temperature much more accurate than a simple "brightness temperature" reading.

Prompt 6: Why LST varies according to the season

The variation of Land Surface Temperature (LST) across seasons is a complex interaction between solar energy, atmospheric conditions, and the physical properties of the earth's surface.¹ In a region like **South 24 Parganas**, these changes are particularly dramatic due to the shift from dry heat to monsoon humidity and finally to winter cooling.

Here are the primary drivers of seasonal LST variation:

1. Solar Zenith Angle and Day Length

The most fundamental cause is the Earth's axial tilt.

- **Summer:** The sun is more directly overhead (closer to a 90° angle). This concentrates solar radiation over a smaller surface area, leading to intense heating.² Additionally, longer daylight hours mean the surface absorbs heat for a longer duration than it can dissipate at night.
- **Winter:** The sun hits the earth at a more oblique angle.³ The energy is spread over a larger area, and shorter days provide less time for the surface to heat up.

2. Soil Moisture and Evaporative Cooling

LST is highly sensitive to how much water is in the soil.⁴

- **Dry Season (Summer):** In South 24 Parganas, many agricultural fields lie fallow and dry. Without moisture, solar energy is converted almost entirely into **Sensible Heat** (the heat you feel), causing LST to spike.
- **Wet Season (Monsoon):** Rain saturates the soil. Solar energy is then used for **Latent Heat** (the energy required to evaporate water) rather than raising the surface temperature. This "evaporative cooling" keeps LST much lower.

3. Vegetation Phenology (The "Greenness" Factor)

Plants act as natural air conditioners for the earth's surface.⁵

- **Summer Stress:** High heat and water scarcity can reduce leaf area index (LAI) in non-mangrove areas. Less leaf cover means more "bare soil" is exposed to direct sunlight, increasing LST.
- **Growth Peaks:** During or after the monsoon, vegetation density increases. Leaves provide shade to the ground and release water vapor through **transpiration**, significantly lowering the surface temperature compared to paved urban areas.

4. Atmospheric Composition and Cloud Cover

The atmosphere acts as a filter and a blanket.

- **Cloud Shading:** In summer and monsoon, heavy cloud cover reflects incoming solar radiation back into space (high albedo), which can temporarily suppress LST.
- **Clear Skies:** In winter, clear skies allow solar radiation to reach the ground easily during the day, but they also allow **Longwave Radiation** (heat) to escape rapidly back into the atmosphere at night, leading to very low nighttime LST.

5. Surface Albedo and Thermal Inertia

Different surfaces react differently to seasonal changes:

- **Urban Surfaces:** Concrete and bitumen have high thermal inertia; they absorb massive amounts of heat in the summer and stay warm long after the sun sets.
- **Water Bodies:** In South 24 Parganas (the land of rivers), water has a high specific heat capacity. It stays cooler than the land in summer and remains slightly warmer than the land in winter, acting as a "seasonal buffer."

Seasonal Summary for South 24 Parganas

Factor	Summer (Pre-Monsoon)	Winter
Solar Intensity	Maximum (Direct rays)	Minimum (Slanted rays)
Primary Cooling	Low (Dry surface)	High (Radiation loss at night)
Surface State	Dry, bare soil hotspots	Moist soil, active cropping
LST Range	32°C to 45°C	15°C to 25°C

Prompt 7: Significance of LST as geophysical parameter

Commissioned Manuscript

Open Access

Satellite Remote Sensing of Global Land Surface Temperature: Definition, Methods, Products, and Applications

[Zhao-Liang Li](#), [Hua Wu](#), [Si-Bo Duan](#), [Wei Zhao](#), [Huazhong Ren](#), [Xiangyang Liu](#), [Pei Leng](#), [Ronglin Tang](#), [Xin Ye](#), [Jinshun Zhu](#), [Yingwei Sun](#), [Menglin Si](#), [Meng Liu](#), [Jiahao Li](#), [Xia Zhang](#), [Guofei Shang](#), [Bo-Hui Tang](#), [Guangjian Yan](#), [Chenghu Zhou](#)

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Land surface temperature (LST) is a crucial geophysical parameter related to surface energy and water balance of the land-atmosphere system. Satellite remote sensing provides the best way to measure LST and generate various LST products at regional and global scales. In this review, to facilitate the application of LST products in different fields, we first present the physical meaning of satellite-derived LST. Subsequently, we summarize recent advances in LST retrieval and validation methods, with a special focus on the state-of-the-art product collections, product accuracies and intercomparisons, and main problems in current LST products as well as their possible solutions. Additionally, we also review the major applications of LST products in agricultural drought monitoring, thermal environment monitoring, thermal anomaly monitoring, and climate change. Finally, we offer recommendations or perspectives to promote LST retrieval methods and their applications. This review will aid the user in gaining a thorough comprehensive understanding of satellite-derived LST products and promoting their appropriate applications.

Prompt 8: Significance of LST in Remote sensing

Land Surface Temperature (LST) is vital in remote sensing as a key indicator of land-atmosphere energy/water exchange, crucial for climate modeling, [agricultural drought monitoring](#), urban heat studies ([UHI](#)), hydrology, and ecology, providing large-scale insights into surface energy balance, vegetation health, and climate change impacts through satellite thermal infrared data. It helps understand land processes, validates climate models, monitors vegetation stress, and assesses environmental issues like urban heat islands.

Key Significance of LST in Remote Sensing:

- **Land-Atmosphere Interaction:** LST reflects how the surface exchanges heat and water with the atmosphere, a fundamental component of the Earth's energy balance.
- **Climate & Weather Studies:** It's used to constrain surface energy budgets, monitor climate change, and improve weather forecasting models.
- **Agriculture & Drought:** Monitoring LST helps detect agricultural drought, assess crop water stress, and manage irrigation.
- **Urban Planning:** Understanding LST is crucial for studying the [Urban Heat Island](#) (UHI) effect, informing urban design and mitigation strategies.
- **Hydrology & Ecology:** It aids in studying water cycles, evapotranspiration, and the health and stress of ecosystems.
- **Environmental Monitoring:** LST helps identify thermal anomalies, monitor environmental changes, and assess land degradation (like barren land).

Remote Sensing's Role:

- Satellites provide the only feasible way to measure LST globally with high spatial and temporal resolution, offering complete area coverage rather than just point measurements.
- Thermal Infrared (TIR) sensors on satellites are used to derive LST, leading to global data products from missions like MODIS, ASTER, and Sentinel.